Assignment 5

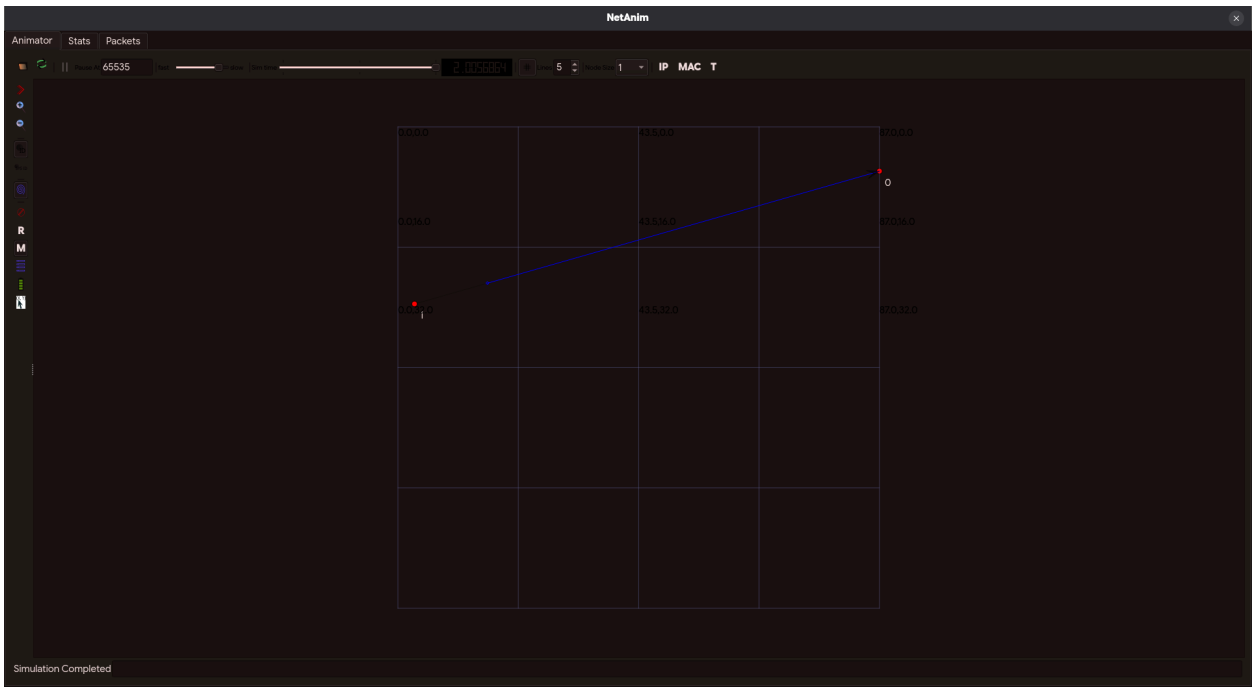
12342390-Rahuldev

Part-1

1)Left red dot → Source node

Right red dot (labeled **0**) → Destination node

0 to 1 stimulation



Explanation:

The red circles represent network nodes.

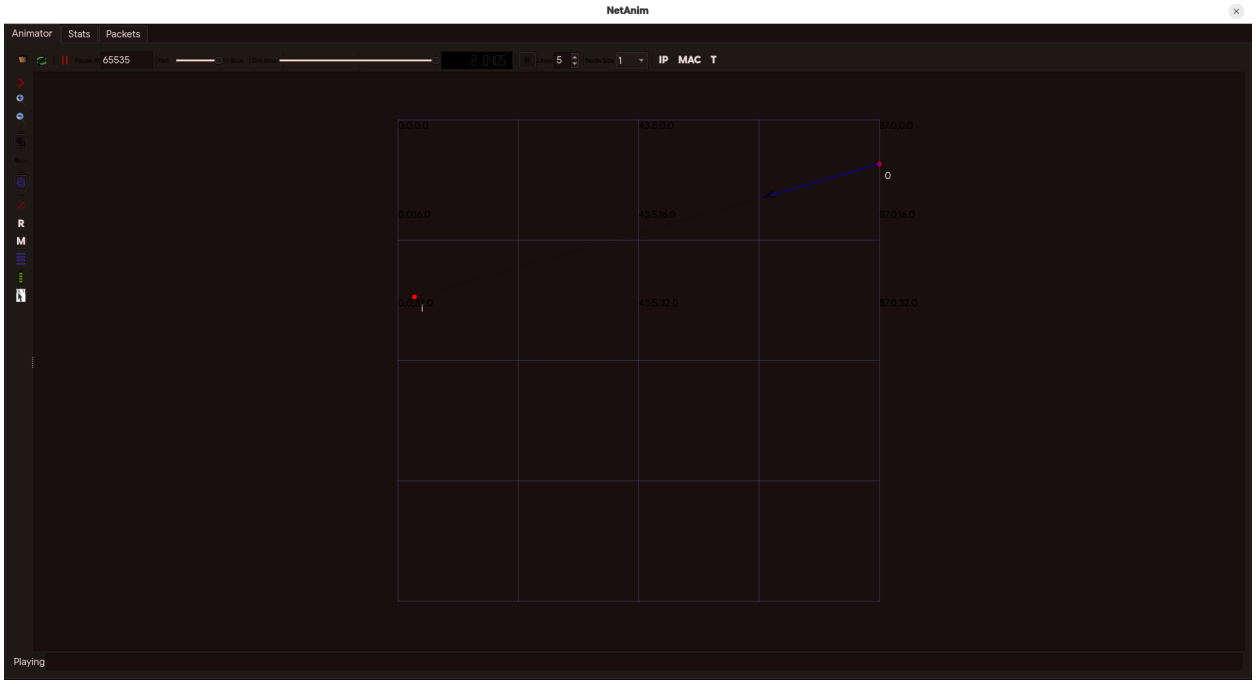
The blue line shows active communication between the nodes.

We can observe packet transmission over the point-to-point link.

The simulation time and controls allow us to analyze network behavior visually

2)Left red node → Source node

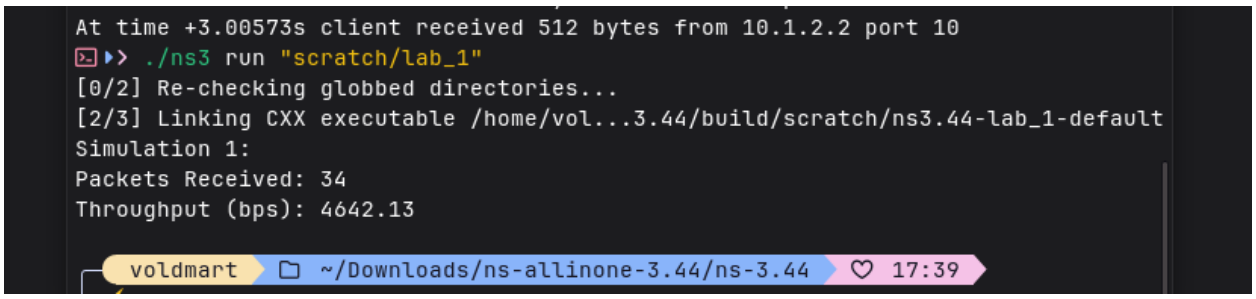
Right red node (labeled “0”) → Destination node
1 to 0 stimulation

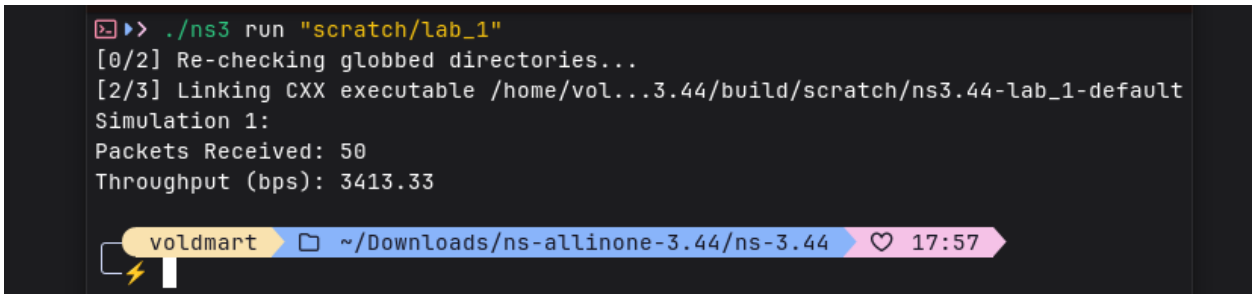


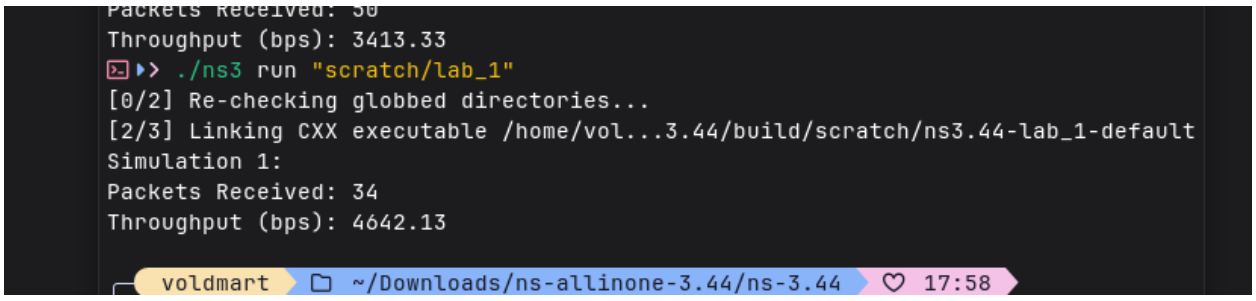
Question 3

Bandwidth	Latency	Maximum Number of Packets	Interval	Packet Size	End-to-End Throughput	Number of packets transmitted
5 Kbps	15 milliseconds	100	1 second	1024	4642.13 bps	34
5 Kbps	10 milliseconds	50	100 milliseconds	512	3413.33 bps	50
5 Kbps	5 milliseconds	Unlimited	10 milliseconds	1024	4642.13 bps	34
5 Mbps	15 milliseconds	150	10 seconds	2048	1638.4 bps	6
5 Mbps	10 milliseconds	50	5 seconds	256	409.6 bps	12

5 Mbps	5 milliseconds	10	1 millisecond	256	341.33 bps	10
5 Gbps	15 milliseconds	Unlimited	1 millisecond	1024	8053420 bps	58985
5 Gbps	20 milliseconds	5	2 seconds	256	170.667 bps	5
5 Gbps	5 milliseconds	100	5 milliseconds	512	6826.67 bps	100







```
At time +3.00573s client received 512 bytes from 10.1.2.2 port 10
 ./ns3 run "scratch/lab_1"
[0/2] Re-checking globbed directories...
[2/3] Linking CXX executable /home/vol...3.44/build/scratch/ns3.44-lab_1-default
Simulation 1:
Packets Received: 34
Throughput (bps): 4642.13

 voldmart  ~/Downloads/ns-allinone-3.44/ns-3.44  17:39
```

```
Command 'build/scratch/ns3.44-lab_1-default' died with <Signals.SIGABRT: 6>.
 ./ns3 run "scratch/lab_1"
[0/2] Re-checking globbed directories...
[2/3] Linking CXX executable /home/vol...3.44/build/scratch/ns3.44-lab_1-default
Simulation 1:
Packets Received: 12
Throughput (bps): 409.6

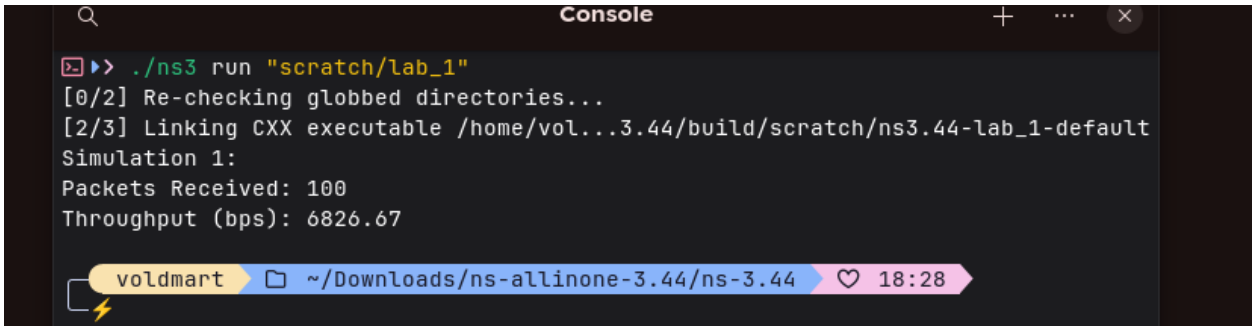
 voldmart  ~/Downloads/ns-allinone-3.44/ns-3.44  18:24
```

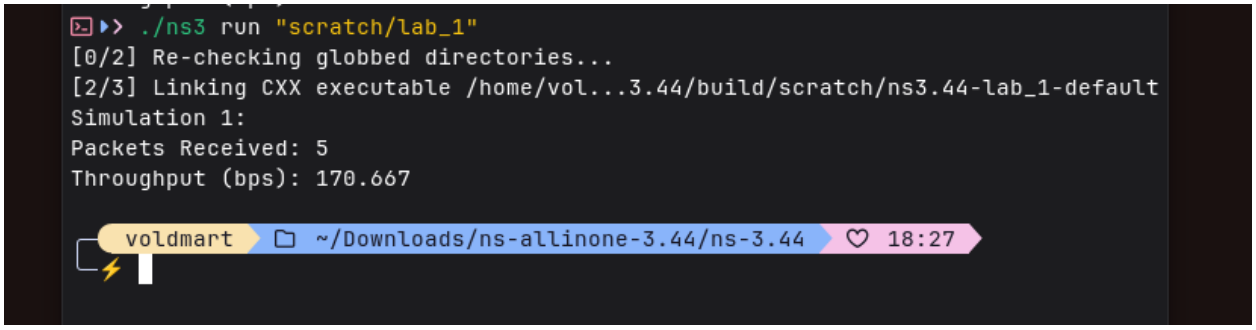
```
Packets Received: 12
Throughput (bps): 409.6
 ./ns3 run "scratch/lab_1"
[0/2] Re-checking globbed directories...
[2/3] Linking CXX executable /home/vol...3.44/build/scratch/ns3.44-lab_1-default
Simulation 1:
Packets Received: 10
Throughput (bps): 341.333

 voldmart  ~/Downloads/ns-allinone-3.44/ns-3.44  18:25
```

```
Console
 ./ns3 run "scratch/lab_1"
[0/2] Re-checking globbed directories...
[2/3] Linking CXX executable /home/vol...3.44/build/scratch/ns3.44-lab_1-default
Simulation 1:
Packets Received: 58985
Throughput (bps): 8.05342e+06

 voldmart  ~/Downloads/ns-allinone-3.44/ns-3.44  18:26
```





Question 4

uint16_t echoPort1 = 9;
uint16_t echoPort2 = 11; // Different port

UdpEchoServerHelper echoServer1(echoPort1);
UdpEchoServerHelper echoServer2(echoPort2);

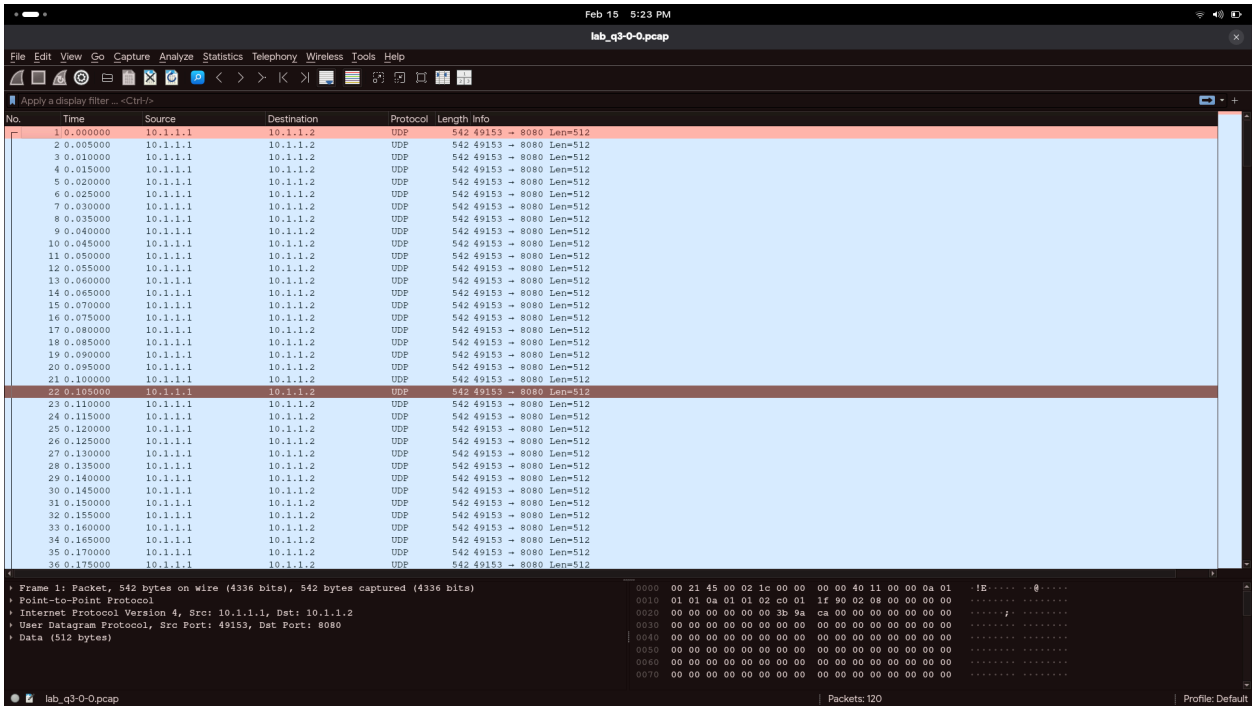
Assign a **different port number** to each `UdpEchoServerHelper`.

Configure each `UdpEchoClientHelper` to send packets to the correct **IP address and port**.

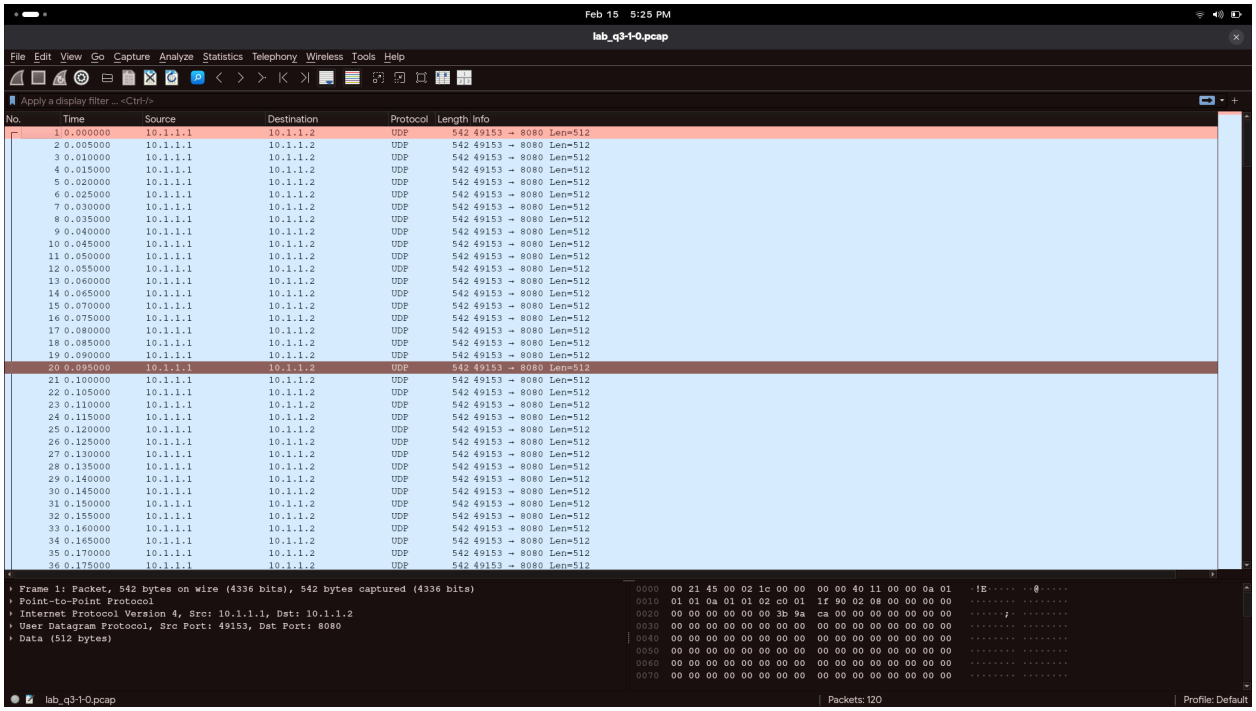
Optionally set different application start times.

Question 5

Node_0 packets



Node 1 packets

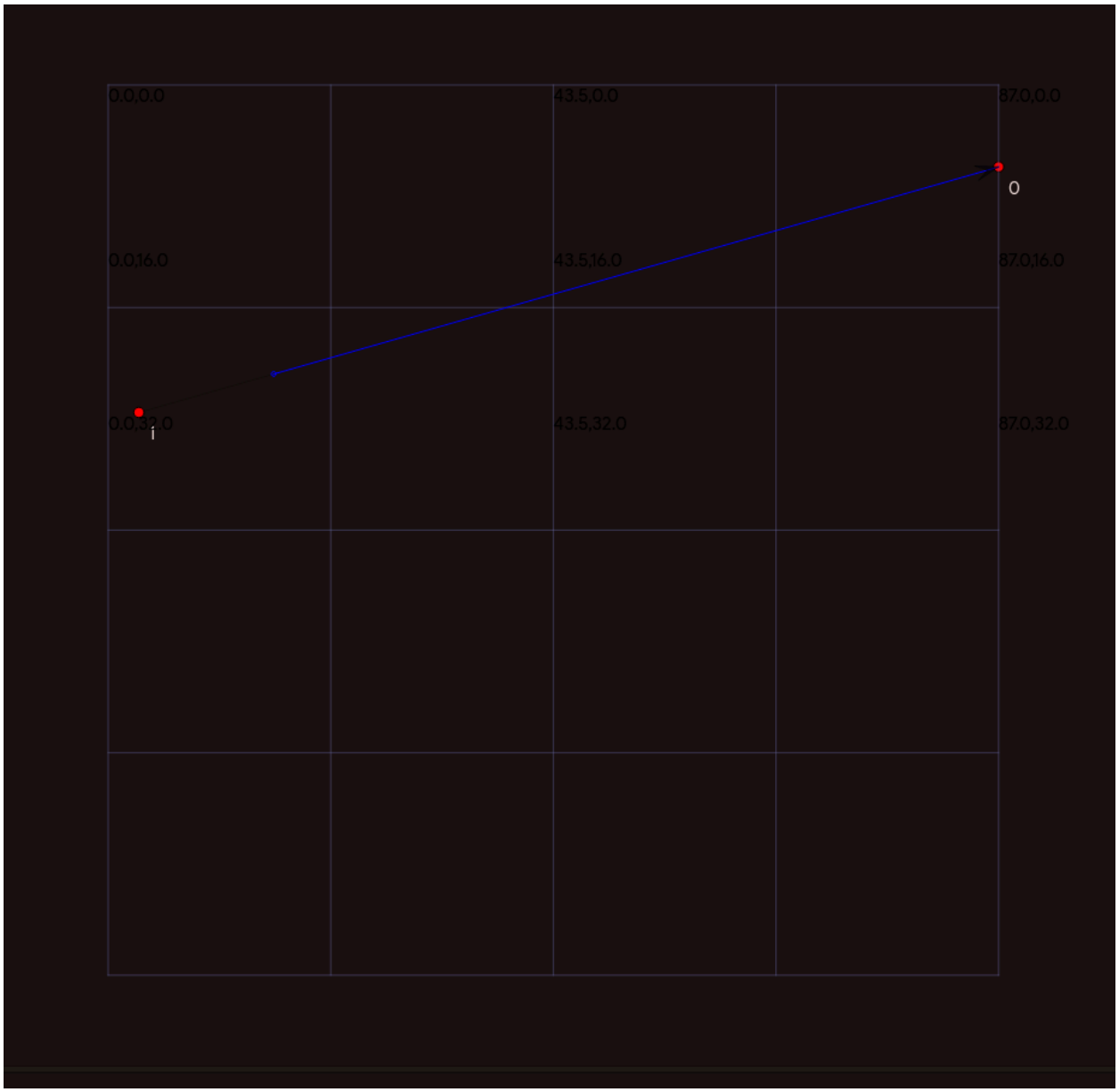


Node0 IP = 10.1.1.1
Node1 Ip = 10.1.1.2

Explanation:

In the packet capture trace, packets labeled DISCARD do not indicate packet loss or an error in communication. Instead, this label shows that the packet was seen at a specific interface or protocol layer, but it was not meant to be processed at that particular point.
During a simulation, a packet may pass through multiple layers such as physical, data link, network, and transport layers. At some of these stages, the packet is only being inspected or forwarded, not consumed. When a layer determines that the packet is not intended for it, the simulation logs the event as DISCARD, meaning not handled here, rather than dropped from the network.
Importantly:
The packet is not lost.The network is not failing.Communication between client and server continues normally.
The UDP packets are still successfully delivered from the client node to the server node.
Therefore, the DISCARD entries simply represent internal packet processing steps within the simulation and do not indicate incorrect delivery or network problems.

Question 6



Explanation

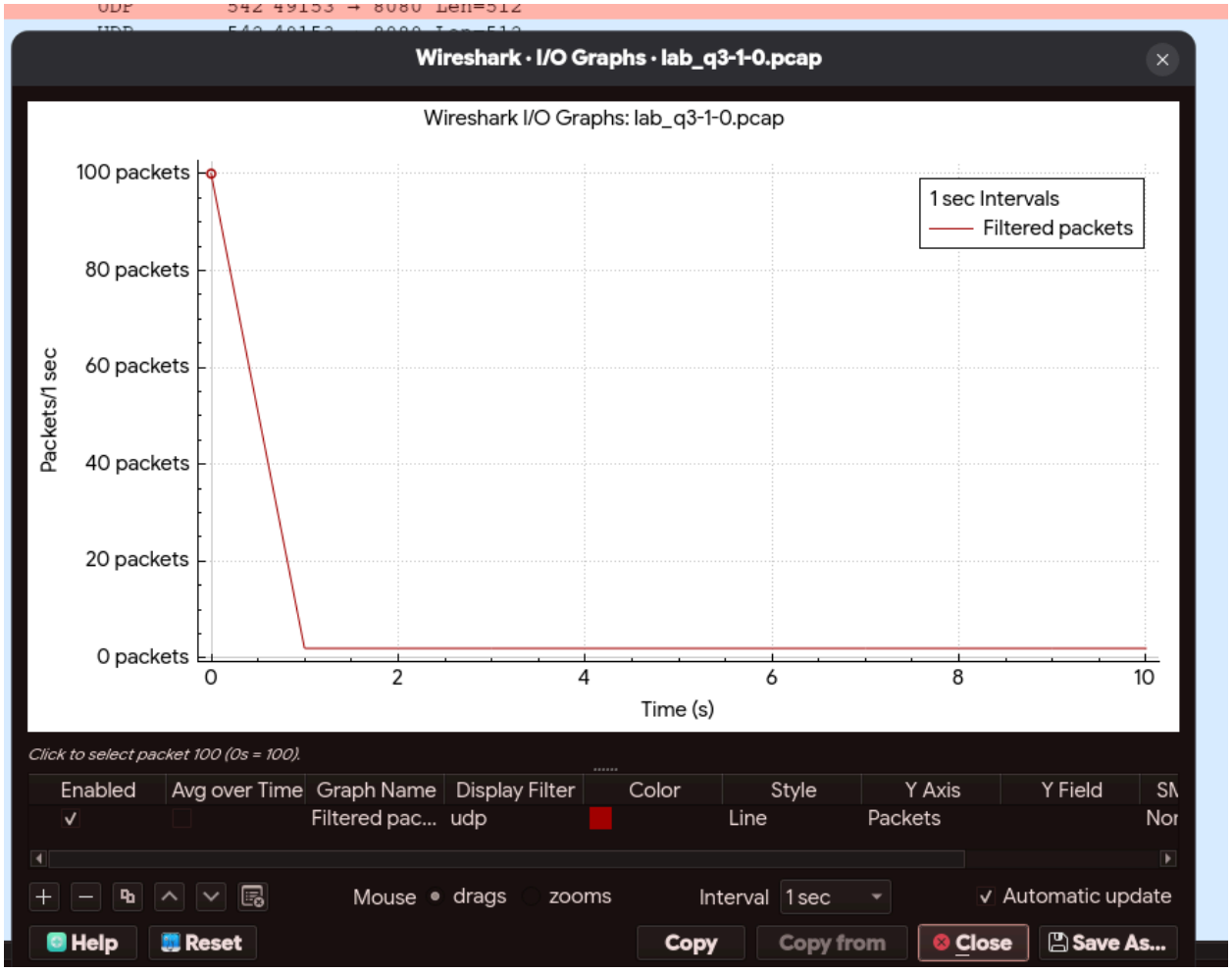
The NetAnim visualization shows the real-time packet flow between the two nodes in the simulation. Node0 functions as the UDP client (source), generating and sending packets, while Node1 acts as the UDP server (destination), receiving those packets. The straight line connecting them represents the point-to-point communication link. The moving packet indicators along this link confirm that data is being transmitted successfully

from the client to the server. The animation visually verifies correct connectivity, proper routing, and successful UDP communication between the nodes during the simulation.

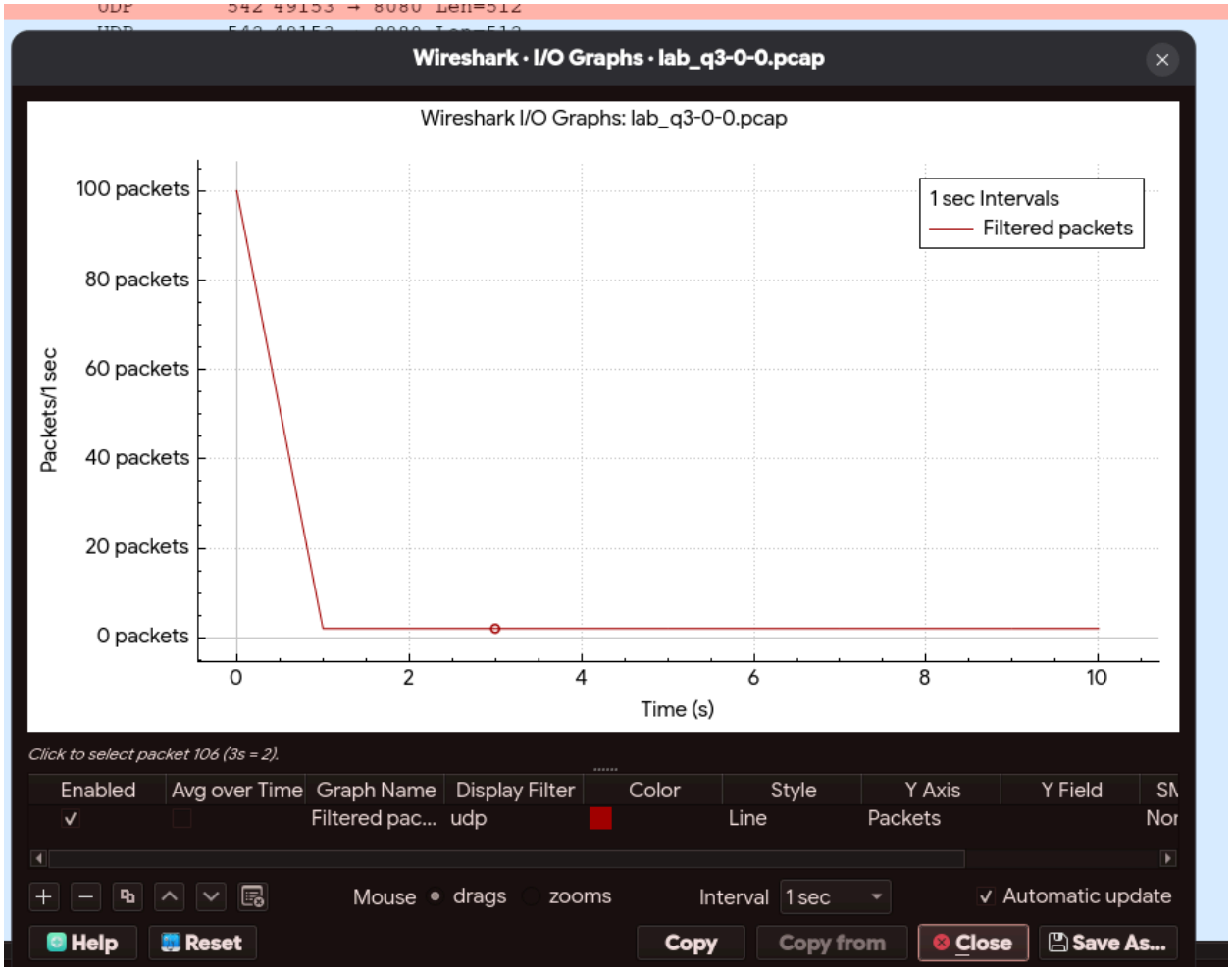
Question 7

The UDP I/O graph shows the number of UDP packets transmitted per second during the simulation. There is a sharp spike at the beginning (around 100 packets/sec in the first second), indicating that the client sent a burst of UDP packets at the start. After this initial transmission, the packet rate quickly drops to nearly zero and remains flat, showing that no further significant UDP traffic occurred. This indicates that the communication was short and burst-based, with all packets sent early in the simulation and no continuous data transfer afterward.

Node 0


UDP 542 49153 → 8080 Len=512
Wireshark · I/O Graphs · lab_q3-1-0.pcap
Wireshark I/O Graphs: lab_q3-1-0.pcap
100 packets
80 packets
60 packets
40 packets
20 packets
0 packets
Packets/1 sec
0
2
4
6
8
10
Time (s)
1 sec Intervals
Filtered packets
Click to select packet 100 (0s = 100).
Enabled
Avg over Time
Graph Name
Display Filter
Color
Style
Y Axis
Y Field
Filtered pac...
udp
Line
Packets
Mouse
drags
zooms
Interval
1 sec
Automatic update
Help
Reset
Copy
Copy from
Close
Save As...

Node 1



Links
Pcapfile 0
https://drive.google.com/file/d/1i4sC3aifQjeeRnMPJiv5arNp44ql0to8/view?usp=sharing
Pcap 1
https://drive.google.com/file/d/1tVdXvQ2q_jMoRCKbIDWQAYjZQWw1n6XL/view?usp=sharing
Code
https://drive.google.com/file/d/1B-4YN3Z42PcWD04Rzuqxc-VJZBeOPRb2/view?usp=drive_link